

Assignment: HDOC Day-16 Activity

1. Problem Statement

Write a menu-driven c program using thw switch-case method to perform digit-based operations on a positive integer. Define one separate user-defined function with a void return type for each menu options without passing parameters. Each function reads the required input, processes it and display the output.

The menu options to be displayed are :

1. Check whether digits are in the increasing order from left to right
2. Check whether digits are in the decreasing order from left to right.
3. Find the largest and smallest digit in the number.
4. Find the second largest and second smallest digit in the number.

2. Algorithm

Main Program:

- **Step 1:** Start.
- **Step 2:** Display menu options (1 to 4).
- **Step 3:** Read user choice.
- **Step 4:** Execute the chosen option using switch:
 - Choice 1: Call checkIncreasing()
 - Choice 2: Call checkDecreasing()
 - Choice 3: Call findMinMax()
 - Choice 4: Call findSecondMinMax()
 - Default: Print "Invalid Choice".
- **Step 5:** Stop.

Function 1: checkIncreasing()

- **Step 1:** Read positive integer num.
- **Step 2:** Extract all digits into an array a[] from right to left.

- **Step 3:** Loop through array to check if digits left-to-right are increasing:
 - If any left digit is not smaller than right digit, mark flag = 0.
- **Step 4:** If flag == 1, print "Increasing order", else print "Not increasing order".

Function 2: checkDecreasing()

- **Step 1:** Read positive integer num.
- **Step 2:** Extract all digits into an array a[] from right to left.
- **Step 3:** Loop through array to check if digits left-to-right are decreasing:
 - If any left digit is not greater than right digit, mark flag = 0.
- **Step 4:** If flag == 1, print "Decreasing order", else print "Not decreasing order".

Function 3: findMinMax()

- **Step 1:** Read positive integer num.
- **Step 2:** Set max = 0 and min = 9.
- **Step 3:** Loop while num > 0:
 - d = num % 10.
 - If d > max, update max = d.
 - If d < min, update min = d.
 - num = num / 10.
- **Step 4:** Print max and min.

Function 4: findSecondMinMax()

- **Step 1:** Read positive integer num.
- **Step 2:** Extract digits into an array a[].
- **Step 3:** Sort the array in ascending order using simple Bubble Sort.
- **Step 4:** Print second element a[1] as second smallest, and second last element a[count - 2] as second largest.

3. TEST CASE TABLE :

TEST CASE	Menu options	input	Expected output	Actual output	status
1	1	1379	Digits are in increasing order	Digits are in increasing order	pass

2	2	8642	Digits are in decreasing order	Digits are in decreasing order	pass
3	3	4815	Largest = 8, Smallest = 1	Largest = 8, Smallest = 1	pass
4	4	2759	2nd Largest = 7, 2nd Smallest = 5	2nd Largest = 7, 2nd Smallest = 5	pass

4. C Program Code

```
#include <stdio.h>
```

```
// Option 1: Check Increasing Order
```

```
void checkIncreasing() {
```

```
    int num, temp, d;
```

```
    int a[20], count = 0, flag = 1;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &num);
```

```
    temp = num;
```

```
    while (temp > 0) {
```

```
        d = temp % 10;
```

```
        a[count] = d;
```

```
        count++;
```

```
        temp = temp / 10;
```

```
    }
```

```

// Check digits from left to right
for (int i = count - 1; i > 0; i--) {
    if (a[i] >= a[i - 1]) {
        flag = 0;
        break;
    }
}

if (flag == 1) {
    printf("The digits of %d are in INCREASING order from left to right.\n", num);
} else {
    printf("The digits of %d are NOT in increasing order from left to right.\n", num);
}
}

```

// Option 2: Check Decreasing Order

```

void checkDecreasing() {
    int num, temp, d;
    int a[20], count = 0, flag = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        d = temp % 10;

```

```
a[count] = d;
count++;
temp = temp / 10;
}
```

```
// Check digits from left to right
```

```
for (int i = count - 1; i > 0; i--) {
    if (a[i] <= a[i - 1]) {
        flag = 0;
        break;
    }
}
```

```
if (flag == 1) {
    printf("The digits of %d are in DECREASING order from left to right.\n", num);
} else {
    printf("The digits of %d are NOT in decreasing order from left to right.\n", num);
}
}
```

```
// Option 3: Find Largest and Smallest
```

```
void findMinMax() {
    int num, temp, d;
    int max = 0, min = 9;

    printf("Enter a positive integer: ");
}
```

```
scanf("%d", &num);

temp = num;
while (temp > 0) {
    d = temp % 10;
    if (d > max) {
        max = d;
    }
    if (d < min) {
        min = d;
    }
    temp = temp / 10;
}

printf("Largest digit: %d\n", max);
printf("Smallest digit: %d\n", min);
}
```

// Option 4: Find Second Largest and Second Smallest

```
void findSecondMinMax() {
    int num, temp, d;
    int a[20], count = 0, t;

    printf("Enter a positive integer: ");
    scanf("%d", &num);
```

```
temp = num;
while (temp > 0) {
    d = temp % 10;
    a[count] = d;
    count++;
    temp = temp / 10;
}
```

```
if (count < 2) {
    printf("Need at least 2 digits.\n");
    return;
}
```

```
// Simple Bubble Sort
```

```
for (int i = 0; i < count - 1; i++) {
    for (int j = 0; j < count - i - 1; j++) {
        if (a[j] > a[j + 1]) {
            t = a[j];
            a[j] = a[j + 1];
            a[j + 1] = t;
        }
    }
}
```

```
printf("Second Smallest digit: %d\n", a[1]);
printf("Second Largest digit: %d\n", a[count - 2]);
```

```
}
```

```
int main() {
```

```
    int choice;
```

```
    printf("--- MENU ---\n");
```

```
    printf("1. Check increasing order from left to right\n");
```

```
    printf("2. Check decreasing order from left to right\n");
```

```
    printf("3. Find largest and smallest digit\n");
```

```
    printf("4. Find second largest and second smallest digit\n");
```

```
    printf("Enter your choice (1-4): ");
```

```
    scanf("%d", &choice);
```

```
    switch (choice) {
```

```
        case 1:
```

```
            checkIncreasing();
```

```
            break;
```

```
        case 2:
```

```
            checkDecreasing();
```

```
            break;
```

```
        case 3:
```

```
            findMinMax();
```

```
            break;
```

```
        case 4:
```

```
            findSecondMinMax();
```

```
            break;
```



```
default:
    printf("Invalid Choice!\n");
}

return 0;
}
```

5. CODE SCREENSHOT AND OUTPUT:

```
#include <stdio.h>

void checkIncreasing() {
    int num, temp, d;
    int a[20], count = 0, flag = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        d = temp % 10;
        a[count] = d;
        count++;
        temp = temp / 10;
    }

    for (int i = count - 1; i > 0; i--) {
        if (a[i] >= a[i - 1]) {
            flag = 0;
            break;
        }
    }

    if (flag == 1) {
        printf("The digits of %d are in INCREASING order from left to right.\n", num);
    } else {
        printf("The digits of %d are NOT in increasing order from left to right.\n", num);
    }
}

void checkDecreasing() {
    int num, temp, d;
    int a[20], count = 0, flag = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        d = temp % 10;
        a[count] = d;
        count++;
        temp = temp / 10;
    }

    for (int i = count - 1; i > 0; i--) {
        if (a[i] <= a[i - 1]) {
            flag = 0;
            break;
        }
    }
}
```

```

    if (flag == 1) {
        printf("The digits of %d are in DECREASING order from left to right.\n", num);
    } else {
        printf("The digits of %d are NOT in decreasing order from left to right.\n", num);
    }
}

void findMinMax() {
    int num, temp, d;
    int max = 0, min = 9;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        d = temp % 10;
        if (d > max) {
            max = d;
        }
        if (d < min) {
            min = d;
        }
        temp = temp / 10;
    }

    printf("Largest digit: %d\n", max);
    printf("Smallest digit: %d\n", min);
}

```

```

void findSecondMinMax() {
    int num, temp, d;
    int a[20], count = 0, t;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    temp = num;
    while (temp > 0) {
        d = temp % 10;
        a[count] = d;
        count++;
        temp = temp / 10;
    }

    if (count < 2) {
        printf("Need at least 2 digits.\n");
        return;
    }

    for (int i = 0; i < count - 1; i++) {
        for (int j = 0; j < count - i - 1; j++) {
            if (a[j] > a[j + 1]) {
                t = a[j];
                a[j] = a[j + 1];
                a[j + 1] = t;
            }
        }
    }
}

```

```

int main() {
    int choice;

    printf("--- MENU ---\n");
    printf("1. Check increasing order from left to right\n");
    printf("2. Check decreasing order from left to right\n");
    printf("3. Find largest and smallest digit\n");
    printf("4. Find second largest and second smallest digit\n");
    printf("Enter your choice (1-4): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            checkIncreasing();
            break;
        case 2:
            checkDecreasing();
            break;
        case 3:
            findMinMax();
            break;
        case 4:
            break;
        case 4:
            findSecondMinMax();
            break;
        default:
            printf("Invalid Choice!\n");
    }

    return 0;
}

```

OUTPUT:

```
Last login: Thu Sep 10 13:04:34 on console
[manalithakar@Manalis-MacBook-Air ~ % nano 16q11.c
[manalithakar@Manalis-MacBook-Air ~ % gcc 16q11.c -o 16q11
[manalithakar@Manalis-MacBook-Air ~ % ./16q11
--- MENU ---
1. Check increasing order from left to right
2. Check decreasing order from left to right
3. Find largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice (1-4): 1
Enter a positive integer: 1379
The digits of 1379 are in INCREASING order from left to right.
[manalithakar@Manalis-MacBook-Air ~ % ./16q11
--- MENU ---
1. Check increasing order from left to right
2. Check decreasing order from left to right
3. Find largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice (1-4): 2
Enter a positive integer: 8642
The digits of 8642 are in DECREASING order from left to right.
[manalithakar@Manalis-MacBook-Air ~ % ./16q11
--- MENU ---
1. Check increasing order from left to right
2. Check decreasing order from left to right
3. Find largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice (1-4): 3
Enter a positive integer: 4815
Largest digit: 8
Smallest digit: 1
[manalithakar@Manalis-MacBook-Air ~ % ./16q11
--- MENU ---
1. Check increasing order from left to right
2. Check decreasing order from left to right
3. Find largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice (1-4): 4
Enter a positive integer: 2759
Second Smallest digit: 5
Second Largest digit: 7
manalithakar@Manalis-MacBook-Air ~ %
```